Custom SCADA Timeline Extension Technical Manual

High-Performance 24-Hour Timeline Control for Siemens WinCC V8 — Version 1.0

> **Validation Status Note:** Tested for browser rendering, dynamic layout scalability, and historical buffer parsing. Runtime execution within Siemens WinCC V8, VBScript performance, and OLEDB SQL bindings should be validated sequentially in your staging environment prior to final commissioning.

1. System Architecture & Core Overview

The Mode Timeline Extension renders a continuous 24-hour industrial process timeline for heavy equipment (e.g., Filter Presses) inside Siemens WinCC V8 Graphics Designer. It uses a hybrid 3-tier ETL architecture to maintain full history persistence across operator screen changes.

Tag Value	Machine Mode Name	Default Color Code
0	Stopped	Red (#DC3545)
1	Starting	Yellow (#FFC107)
2	Working	Green (#28A745)
3	Stopping	Orange (#FD7E14)

2. Repository & Package Contents

The bundle includes the custom web control package, VBScript extraction scripts, and documentation assets.

Path / Component	Description
`{92DF087E-5FE1-4828-B1DB-40C868DA8D0F}.zip`	Compiled Web Control (Do not rename folder/ZIP name)
`vbs-scripts/OnOpen_Timeline_ETL.vbs`	ETL Script for Graphics Designer OnOpen event

3. WinCC V8 Installation Steps

Follow these precise steps to deploy the extension safely into your existing WinCC V8 runtime environment:

- **Step 1:** Navigate to your project directory and locate the `GraCS` folder. Create a subfolder named `CustomWebControls` if it does not already exist.
- **Step 2:** Extract or copy the control ZIP file containing its precise GUID filename directly into `GraCS\CustomWebControls`.
- **Step 3:** Remove any legacy versions or uncontracted controls from alternative paths to avoid version caching collisions.
- **Step 4:** Restart Siemens WinCC Graphics Designer to refresh the control cache completely.
- **Step 5:** Drag the **"Mode Timeline"** control from the Web Controls palette onto your target engineering picture.
- **Step 6:** Bind properties to your live PLC process tag (`status`) and internal text tags (`modeData1` to `modeData4`).

4. Mandatory Property Bindings

Property Name	Target WinCC Tag Binding
`status`	Live PLC integer tag (e.g., `Machine_Mode`, values 0–3)
`modeData1` … `modeData4`	Internal text tags (length set to 252 characters each)

5. Troubleshooting Matrix

If the timeline fails to load or shows warnings during startup, verify character length configurations on text tags and check the WinCC SQL database catalog connection string inside the VBScript file.